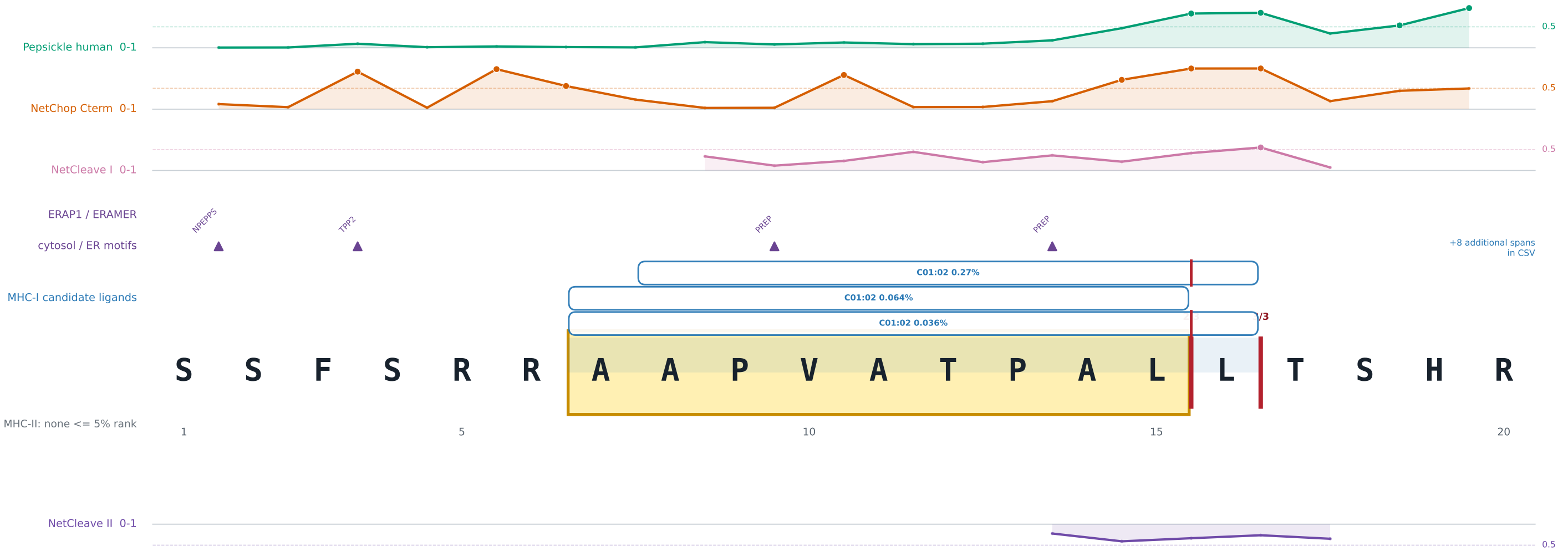


PIP5K1A p.Gly474fs| JLF V3 | 20 aa

disclosed minimal epitope AAPVATPAL at 7-15

INTRACELLULAR / CLASS-I PROCESSING - red slash = 2/3 or 3/3 displayed tracks at >=0.5



SERUM / EXTRACELLULAR - native scores and recognition motifs



Reading the map. Each profile joins exact scores at adjacent assessed bonds; it is not smoothed, averaged, or calibrated, and gaps are unassessed. Dots at or beyond the dashed 0.5 line are display candidates. A red slash is drawn only when all three displayed intracellular tracks assess that bond and >=2 score >=0.5; its label is the supporting fraction (2/3 or 3/3), not a probability. Pale blue/purple sequence tint marks coverage by a displayed MHC window; outlined windows identify the individual candidate MHC ligands at <=2%/<=5% rank (top-ranked non-overlapping spans shown; all rows are in the CSV); red ticks inside them are internal candidate cuts before binding. These counts are not probabilities, and ligand bars do not model binding occupancy, timing, or post-binding protection.